

Lab Assignment 3: Time-Series Regression: Stock Price Prediction

CSAI2017P: Applied Machine Learning (Lab)

Experiment 4 (Unit V) • CO2, CO4

Instructor: Dr. Tofik Ali**Due:** announced in the lab**Student Name:** _____**SAP ID:** _____**Batch:** _____**Lab quiz:** LQ3, after submission

Objective

- Build supervised features from an ordered series without leaking the future.
- Split time-ordered data correctly and see why a random split is wrong.
- Compare a model against the naive baseline that actually matters.

Datasets used in this assignment

C1 (daily stock OHLCV).

Full descriptions, sources and loading snippets for all anchor datasets are in **Anchor-Datasets.pdf**, alongside this brief.

Instructions

- Use a train/test split (80/20 unless stated) and set `random_state` for reproducibility.
- Fit every transformer inside a `Pipeline` so nothing is fitted on the test set.
- Report the metric named in the question. A number without units or context earns no marks.
- Every question asks for a short written observation. Code without commentary caps you at half marks.
- The lab quiz that follows will ask you to read, trace and modify *your own* submitted code.

Submit:

- Notebook: `AML_LA03_<SAPID>.ipynb` with all cells executed and outputs visible.
- Report: 2–3 pages PDF, or a `README.md`, carrying your results table and observations.
- Do not upload datasets. Cite the source and include the loading code.

Section	Level	Choice rule	Marks
A	Easy	Attempt any 2 of 3 (2 marks each)	4
B	Medium	Attempt any 1 of 2 (4 marks each)	4
C	Hard	Compulsory	2
Total			10

Course outcomes assessed by this assignment: **CO2, CO4**. Attempting more than the choice rule allows is not penalised; the best attempts are marked.

Section A (Easy) — attempt any 2 of 3 ($2 \times 2 = 4$)**A1. Load and describe the series** [2 marks]

Dataset: C1, one ticker, ≥ 3 years.

- (a) Load the daily history, parse the date column and sort ascending.
- (b) Plot the closing price and state the date range and number of trading days.
- (c) State in two lines why the row order matters here but did not in LA02.

A2. Lag features [2 marks]

Dataset: C1.

- (a) Create features `Close_lag1`, `Close_lag2`, `Close_lag3` and a 5-day rolling mean.
- (b) Set the target to the next day's close. Show the first five rows of the resulting frame.
- (c) Explain in two lines why the first few rows must be dropped.

A3. The naive baseline [2 marks]

Dataset: C1.

- (a) Build the baseline that predicts tomorrow's close equals today's close.
- (b) Report its MAE and RMSE on your test period.
- (c) State in two lines why any model must beat this number to be interesting.

Section B (Medium) — attempt any 1 of 2 ($1 \times 4 = 4$)**B1. Random split versus chronological split** [4 marks]

Dataset: C1.

- (a) Train a `RandomForestRegressor` twice: once with `train_test_split(shuffle=True)`, once with the last 20% of dates held out.
- (b) Report test MAE for both and put them side by side.
- (c) Explain in 6–8 lines why the shuffled number is meaningless, naming the mechanism.

B2. Horizon and difficulty [4 marks]

Dataset: C1, chronological split throughout.

- (a) Retrain your model to predict the close 1, 5 and 10 trading days ahead.
- (b) Report MAE for each horizon alongside the persistence baseline at the same horizon.
- (c) Plot MAE against horizon and explain the trend in 4–6 lines.

Section C (Hard) — compulsory (2)**C1. Would you trade on it?** [2 marks]

Dataset: C1.

- (a) Convert your best model's predictions into an up/down direction call and report directional accuracy.

- (b) In 4–6 lines, state whether a low MAE with near-50% directional accuracy is useful, and why.

End of Lab Assignment 3

Sources: Müller and Guido, *Introduction to Machine Learning with Python*, companion notebooks.;McKinney, *Python for Data Analysis* (3e), O'Reilly, 2022.;Scikit-learn user guide, accessed 2026-08-04.;Matplotlib and Seaborn documentation, accessed 2026-08-04.